Artificial Neural Networks and Deep Architectures, DD2437

Short report on lab assignment 1a

Classification with a single-layer perceptron

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1 Main objectives and scope of the assignment

Our major goals in this assignment were:

- to implement a single-layer perceptron for binary classification tasks,
- to apply and compare perceptron learning and the Widrow–Hoff (delta) learning rule using sequential and batch training,
- to investigate the effects of linear separability, bias, and data sampling on convergence and classification performance.

The scope of this lab is limited to two-dimensional synthetic datasets and single-layer networks. The analysis focuses on convergence behavior and generalization rather than optimal performance. It is assumed that the data distributions are known and generated artificially, and that performance is evaluated only on the available samples.

2 Methods

All experiments were implemented in Python. NumPy was used for data generation, matrix operations, and implementation of the perceptron learning rule and the Widrow–Hoff (delta) rule. Matplotlib was used to visualize the datasets, decision boundaries, and learning curves.

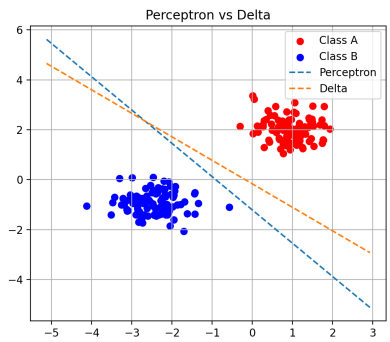
Both sequential (online) and batch learning schemes were implemented. Model performance was evaluated using classification accuracy and mean squared error computed over each training epoch.

3 Results and discussion

3.1 Classification with a single-layer perceptron

Comparison of perceptron and delta learning

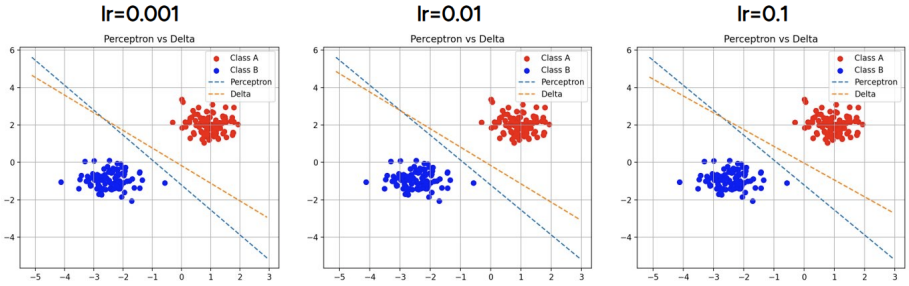
Two linearly separable classes were generated from multivariate normal distributions in a two-dimensional space. The dataset was chosen such that a linear decision boundary exists, allowing both learning algorithms to converge to a correct solution. Both classical perceptron learning and the Widrow–Hoff (delta) rule successfully solved the classification task and converged to a separating decision boundary. As shown in Figure 1, the decision boundary obtained using the delta rule is approximately equidistant from the two classes, resulting in a more balanced separation. In contrast, the classical perceptron solution places one training sample very close to the decision boundary, indicating a less optimal margin despite correct classification.



Figur 1: Decision boundaries obtained using classical perceptron learning and the Widrow–Hoff (delta) rule on the linearly separable dataset.

Effect of learning rate

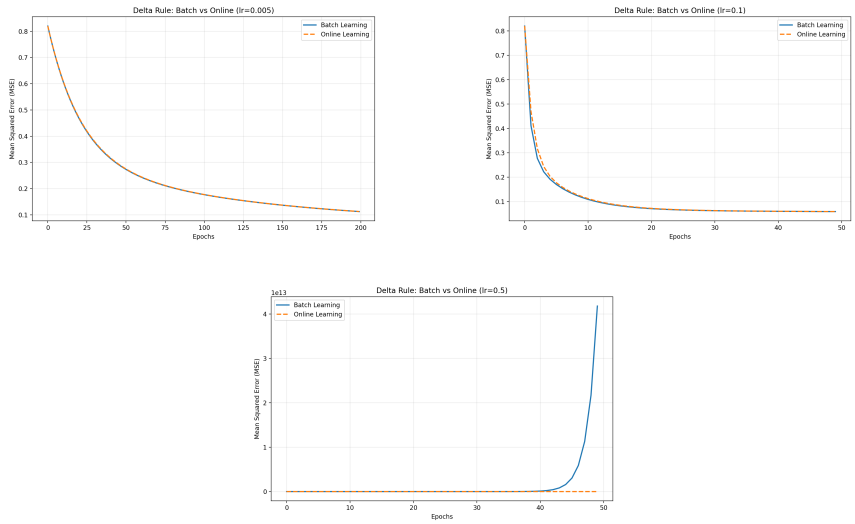
Figure 2 shows the effect of varying the learning rate on convergence and classification performance. For classical perceptron learning, changes in the learning rate did not affect the final classification result, provided that convergence was achieved. For the delta rule, the learning rate influenced the speed of convergence, but all tested values led to successful convergence. The transition between different learning rates was smooth, with larger learning rates resulting in faster convergence and smaller learning rates leading to slower but more stable updates. Overall, the delta rule showed robust and stable behavior with respect to learning rate variations in this experiment.



Figur 2: Effect of different learning rates on convergence and classification performance for classical perceptron learning and the Widrow–Hoff (delta) rule.

Sequential versus batch delta learning

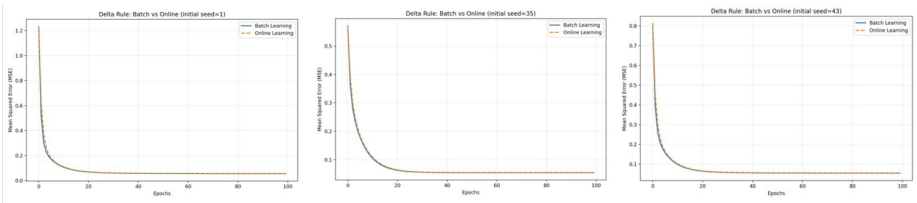
Sequential (online) and batch implementations of the delta rule were compared by analyzing learning curves for different learning rates, as shown in Figure 3. Due to the simplicity of the task, both approaches converged to similar solutions. However, the sequential method converged faster during the initial epochs, particularly for moderate learning rates (e.g. $\eta = 0.1$). Increasing the learning rate generally accelerated convergence, but for large values (e.g. $\eta = 0.5$), batch learning exhibited overshooting effects, leading to large weight updates and unstable behavior.



Figur 3: Comparison of sequential and batch implementations of the delta rule for different learning rates: $\eta = 0.005$ (top left), $\eta = 0.1$ (top right), and $\eta = 0.5$ (bottom).

Sensitivity to random initialization

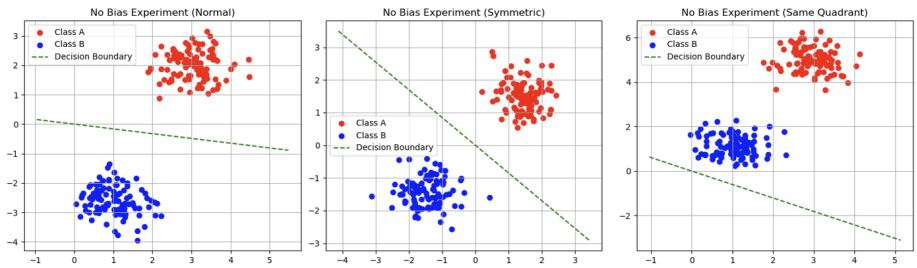
The influence of random weight initialization was investigated by training the network multiple times with different initial conditions. Figure 4 shows that different initializations resulted in varying convergence speeds. However, the final classification performance and decision boundaries remained similar across runs, indicating low sensitivity to initialization for this experiment.



Figur 4: Effect of different random weight initializations on convergence.

Effect of removing the bias term

The effect of removing the bias term was studied using batch delta learning. Figure 5 demonstrates that correct classification without a bias term is only possible for specific data configurations. When both classes are symmetrically distributed with respect to the axes or the origin, or when samples occupy different quadrants, the perceptron can converge without bias. In contrast, when both classes lie within the same quadrant or collectively span three quadrants, correct classification is not possible without a bias term.



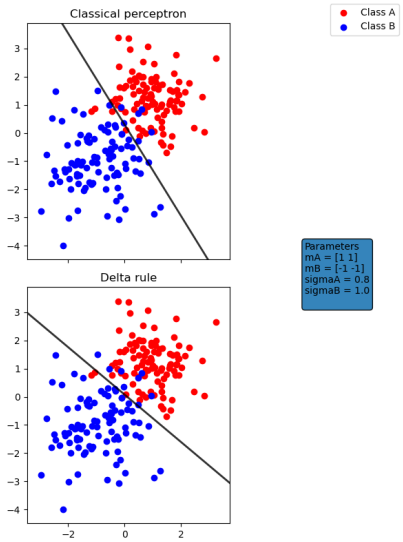
Figur 5: Classification results obtained using batch delta learning with and without a bias term.

3.2 Classification of data that are not linearly separable

Overlapping and non-linearly separable data

To analyze the behavior of a single-layer perceptron on non-linearly separable data, the class distributions were generated such that significant overlap existed

between the two classes. Figure 6 shows that, due to this overlap, no linear decision boundary can correctly classify all samples.



Figur 6: Non-linearly separable dataset with overlapping class distributions.

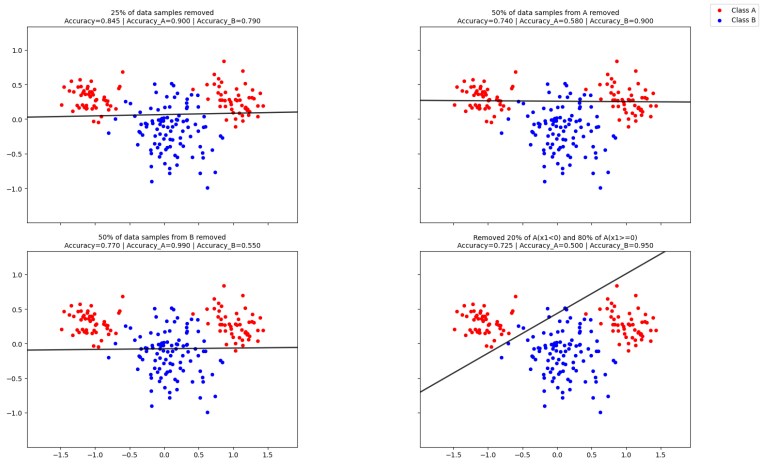
As expected, neither classical perceptron learning nor the Widrow–Hoff (delta) rule was able to achieve perfect classification. For both methods, a subset of samples from each class remained misclassified regardless of training duration. This reflects the inherent ambiguity of the dataset, where certain regions of the input space cannot be unambiguously assigned to a single class.

A comparison of class-wise performance revealed that the delta rule tended to classify samples from class A more accurately, whereas classical perceptron learning achieved better classification performance for class B. However, due to the overlapping nature of the data, it is difficult to determine which method performs better overall. In regions where the class distributions overlap, misclassifications are unavoidable, and differences in performance are strongly dependent on how each learning rule balances errors between the two classes.

Effect of data subsampling

The resulting decision boundaries of the subsampling cases are shown in Figure 7.

When samples were removed randomly from both classes, the learned decision boundary remained relatively similar to that obtained using the full dataset, although increased variability was observed across different runs. In contrast, removing a large fraction of samples from only one class introduced a strong bias in the learned boundary, which shifted toward the underrepresented class and significantly degraded its classification performance.



Figur 7: Decision boundaries obtained using batch delta learning under different subsampling strategies.

The most pronounced effects were observed in the case of region-based subsampling. By removing samples from specific areas of the input space, the classifier learned a distorted representation of the data distribution, leading to poor classification performance in the removed regions. These results demonstrate that the perceptron trained with the delta rule learns the distribution of the available training data rather than the true underlying distribution, and that sampling bias has a strong negative impact on generalization.

4 Final remarks

The results demonstrate that learning outcomes are strongly influenced by data characteristics such as separability, balance, and representativeness. Sampling bias and class imbalance can significantly distort the learned decision boundary and lead to misleading performance estimates, even when the training error appears low. These effects are particularly evident for non-linearly separable data, where ambiguity in the input space makes perfect classification impossible.

Clear differences between classical perceptron learning and the Widrow–Hoff (delta) rule were observed. While both methods perform well on linearly separable data, the delta rule produces more stable and consistent solutions. The learning rate primarily affects the speed of convergence rather than the final classification outcome. In particular, the delta rule showed smooth and robust behavior across different learning rates, whereas the final result of classical perceptron learning remained largely unchanged once convergence was achieved. Overall, the experiments highlight the importance of both data quality and learning dynamics in determining reliable generalization.